Posterior Sampling Time Exploration – $m = 64$ and $T = 500$

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The overall model fitting time (with `equalTimeDist = TRUE` specified when fitting the models) for our 4 models, i.e., `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`, are 7.04 hours, 7.03 hours, 7.04 hours, and 6.31 hours, respectively.[1] If we do not take advantage of our tactics for evenly dispersed time points presented in Appendix B of our manuscript by specifying `equalTimeDist = FALSE` instead, we will need around 27 days to fit the same 4 models using the same computation resources, as $T = 500$ is relatively large. This corresponds well to what we have discussed in Appendix B regarding our approaches' manifest computational acceleration in Gibbs sampler steps for temporal parameters ψ, Υ, and $\boldsymbol{\eta}_t$'s. Since $m = 64$ is quite small, there aren't any significant differences in the recorded Gibbs sampler time between our four models, as expected.

We first display the first 50 kept post-burn-in MCMC iterations' posterior sampling time (in milliseconds) for 10 key Gibbs sampler steps (corresponding to ψ, Υ, $\boldsymbol{\eta}_t$'s, $z_{jl}(\boldsymbol{s}_{i,o})$'s or $u_j(\boldsymbol{s}_{i,o})$'s, $\xi_j(\boldsymbol{s}_{i,o})$'s, θ_{jl_j}'s, $\delta_{1:k}$, ρ, κ, and $\alpha_{jl_j}(\boldsymbol{s}_{i,o})$'s) for our 4 models, i.e., `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`.

```r
setwd(wd)
load("GibbsStepTimeFixedLfullGPfast.RData")
load("GibbsStepTimeFixedLblockFast.RData")
load("GibbsStepTimeFixedLsequenFast.RData")
load("GibbsStepTimeVaryLjSequenFast.RData")
head(GibbsStepTimeFixedLfullGP.fast, 50)
```

```
##        z xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 82 32   317     1     3     1   2 327       2  53
##  [2,] 83 34   307     1     3     1   1 330       2  54
##  [3,] 81 32   306     1     3     1   1 322       2  53
##  [4,] 82 33   305     1     3     1   1 323       2  54
##  [5,] 83 34   309     1     3     1   1 324       2  53
##  [6,] 81 33   318     1     3     1   1 329       2  53
##  [7,] 83 33   326     1     3     1   1 329       2  54
##  [8,] 85 32   319     1     3     1   1 327       2  56
##  [9,] 82 33   315     1     3     1   1 325       2  54
## [10,] 84 32   311     1     3     1   2 323       2  54
## [11,] 85 34   323     1     3     1   1 326       2  53
## [12,] 84 34   316     1     3     1   1 323       2  54
## [13,] 87 34   320     1     3     1   1 327       2  53
## [14,] 83 33   321     1     3     1   1 328       2  53
## [15,] 87 32   334     1     3     1   1 327       2  53
## [16,] 84 34   319     1     3     1   2 328       2  53
## [17,] 84 33   328     1     3     1   2 322       2  52
## [18,] 84 32   316     1     3     1   1 319       2  52
## [19,] 82 34   314     1     3     1   1 325       2  53
## [20,] 83 34   311     1     3     1   2 327       2  54
```

[1] It is worth noting that we utilized a version of our package with the extra step-wise time recording code portions to fit the above models. This package version is significantly slower than our original package version presented in the GitHub repository.

```
## [21,] 82 33   308     1     3     1   1 324       2  55
## [22,] 82 33   310     1     3     1   2 349       2  55
## [23,] 82 33   333     1     3     1   1 333       2  57
## [24,] 84 34   324     1     3     2   2 325       2  54
## [25,] 82 35   353     1     3     1   2 331       2  53
## [26,] 84 34   324     1     3     1   2 328       2  54
## [27,] 81 33   331     1     3     1   1 325       2  52
## [28,] 83 34   325     1     3     1   2 323       2  52
## [29,] 84 34   318     1     3     1   2 321       2  51
## [30,] 80 34   317     1     3     1   1 323       2  52
## [31,] 82 33   317     1     3     1   2 316       2  54
## [32,] 83 33   317     1     3     1   2 327       2  56
## [33,] 77 33   314     1     3     1   1 326       2  56
## [34,] 83 34   318     1     3     1   2 342       2  57
## [35,] 83 32   323     1     3     1   1 335       2  54
## [36,] 81 33   325     1     2     1   1 330       2  54
## [37,] 81 33   330     1     3     1   2 329       2  55
## [38,] 83 34   327     1     3     1   1 327       2  57
## [39,] 81 32   323     1     3     1   1 323       2  55
## [40,] 83 34   322     1     3     1   1 325       2  55
## [41,] 83 34   319     1     3     1   1 326       2  57
## [42,] 81 34   316     1     3     1   1 329       2  57
## [43,] 84 35   320     1     3     1   1 323       2  56
## [44,] 84 33   321     1     3     1   1 331       2  57
## [45,] 82 32   317     1     3     1   1 338       2  55
## [46,] 84 34   311     1     3     1   1 326       2  53
## [47,] 86 34   319     1     3     1   1 349       2  58
## [48,] 83 32   312     1     3     1   1 338       2  55
## [49,] 82 33   322     1     3     1   1 334       2  56
## [50,] 87 34   342     1     3     1   1 343       2  58
```

```r
head(GibbsStepTimeFixedLblock.fast, 50)
```

```
##         z xi theta delta alpha kappa rho eta upsilon psi
##  [1,]  82 35   325     1     3     1   1 324       2  52
##  [2,]  85 34   322     1     3     1   1 324       2  51
##  [3,]  82 33   314     1     3     1   1 313       2  51
##  [4,]  83 35   317     1     3     1   2 315       2  52
##  [5,]  81 32   318     1     3     1   1 323       2  52
##  [6,]  86 34   313     1     3     1   1 330       2  54
##  [7,]  84 35   307     1     3     1   1 327       2  53
##  [8,]  81 33   312     1     3     1   1 329       2  54
##  [9,]  83 34   314     1     3     1   1 325       2  55
## [10,]  84 32   315     1     3     1   1 326       2  54
## [11,]  81 35   321     1     3     1   1 324       2  54
## [12,]  84 32   322     1     3     1   1 330       2  53
## [13,]  83 32   315     1     3     1   1 324       2  54
## [14,]  85 33   319     1     3     1   1 321       2  53
## [15,]  85 31   315     1     3     1   1 321       2  52
## [16,]  82 33   312     1     3     1   1 316       2  53
## [17,]  82 32   309     1     3     1   1 326       2  52
## [18,]  82 32   317     1     3     1   1 328       2  52
## [19,]  83 32   324     1     3     1   1 322       2  52
## [20,]  84 33   317     1     3     1   1 330       2  53
## [21,]  84 34   330     1     3     1   1 332       2  55
```

```
## [22,]  85 32   314     1     3     1   1 329       2  54
## [23,]  84 33   319     1     3     1   1 324       2  54
## [24,]  87 32   310     1     3     1   1 334       2  54
## [25,]  86 33   310     1     3     1   1 335       2  56
## [26,]  83 33   315     1     3     1   1 336       2  53
## [27,]  85 33   321     1     3     1   1 347       2  55
## [28,]  84 33   322     1     3     1   1 328       2  53
## [29,]  83 32   324     1     3     1   1 329       2  53
## [30,]  90 34   321     1     3     1   1 325       2  60
## [31,]  89 33   340     1     3     1   1 325       2  52
## [32,]  82 33   316     1     3     1   1 314       2  52
## [33,]  82 33   317     1     3     1   1 325       2  51
## [34,]  85 34   313     1     3     1   1 334       2  53
## [35,]  81 33   314     1     3     1   1 339       2  55
## [36,]  81 32   306     1     3     1   1 333       2  55
## [37,]  81 32   315     1     3     1   1 330       2  54
## [38,]  82 32   314     1     3     1   1 328       2  54
## [39,]  82 32   311     1     3     1   1 328       2  56
## [40,]  80 33   318     1     3     1   1 326       2  54
## [41,]  82 32   317     1     3     1   1 327       2  55
## [42,]  87 32   315     1     3     1   1 320       2  53
## [43,]  80 33   318     1     3     1   1 321       2  53
## [44,]  82 33   319     1     3     1   1 325       2  55
## [45,]  82 33   318     1     3     1   1 325       2  57
## [46,]  82 32   318     1     3     1   1 329       2  55
## [47,]  82 34   318     1     3     1   1 324       2  54
## [48,] 218 34   318     1     3     1   1 318       2  53
## [49,]  81 34   318     1     3     1   1 321       2  55
## [50,]  85 33   313     1     3     1   1 317       2  54
```

```r
head(GibbsStepTimeFixedLsequen.fast, 50)
```

```
##        z xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 75 33   321     1     9     1   1 327       2  57
##  [2,] 77 34   318     1     9     1   1 325       2  57
##  [3,] 80 33   322     1     9     1   1 329       2  56
##  [4,] 73 33   316     1    12     4   3 336       6  60
##  [5,] 77 32   316     1     9     1   1 331       2  56
##  [6,] 77 34   319     1     9     2   1 328       2  53
##  [7,] 76 32   327     1     9     1   1 330       2  54
##  [8,] 76 33   322     1     9     1   1 324       2  53
##  [9,] 79 34   321     1     9     1   1 326       2  53
## [10,] 79 33   328     1     9     1   1 328       2  54
## [11,] 74 32   319     1     9     1   1 323       2  54
## [12,] 76 34   320     1     9     1   1 326       2  53
## [13,] 76 34   323     1     9     1   1 323       2  53
## [14,] 78 34   327     1     9     1   1 323       2  53
## [15,] 79 35   319     1     9     1   2 321       2  53
## [16,] 78 32   320     1     9     1   1 339       2  53
## [17,] 76 32   318     1     9     1   1 332       2  55
## [18,] 78 34   315     1     9     1   2 332       2  54
## [19,] 75 33   311     1     9     1   2 328       2  53
## [20,] 75 32   323     1     9     1   1 330       2  53
## [21,] 77 32   329     1     9     1   2 341       2  54
## [22,] 75 32   329     1     9     1   2 329       2  53
```

```
## [23,] 81 33   322     1     9     1   2 325       2  54
## [24,] 75 32   321     1     9     1   1 322       2  56
## [25,] 77 34   323     1     9     1   1 324       2  53
## [26,] 76 34   324     1     9     1   1 325       2  54
## [27,] 76 33   320     1     9     1   2 327       2  53
## [28,] 78 33   322     1     9     1   1 318       2  54
## [29,] 76 32   309     1     9     1   1 335       2  55
## [30,] 75 32   311     1     9     1   2 326       2  53
## [31,] 77 34   316     1     9     1   2 331       2  55
## [32,] 75 33   316     1     9     1   1 331       2  55
## [33,] 78 33   320     1     9     1   1 332       2  55
## [34,] 76 34   330     1     9     1   2 334       2  56
## [35,] 76 33   320     1     9     1   1 325       2  55
## [36,] 78 33   321     1     9     1   1 326       2  57
## [37,] 78 34   326     1     9     1   2 331       2  57
## [38,] 78 34   322     1     9     1   1 333       2  57
## [39,] 78 36   332     1     9     1   1 330       2  55
## [40,] 76 33   317     1     9     1   2 323       2  56
## [41,] 82 33   324     1     9     1   1 320       2  55
## [42,] 78 35   323     1     9     1   2 322       2  56
## [43,] 76 32   314     1     9     1   2 328       2  54
## [44,] 80 34   318     1     9     1   1 330       2  53
## [45,] 76 32   313     1     9     1   1 331       2  53
## [46,] 80 35   317     1     9     1   2 330       2  54
## [47,] 77 33   314     1     9     1   1 327       2  54
## [48,] 77 34   320     1     9     1   2 332       2  53
## [49,] 79 33   317     1     9     1   1 333       2  54
## [50,] 76 32   320     1     9     1   2 333       2  53
```

```r
head(GibbsStepTimeVaryLjSequen.fast, 50)
```

```
##       u xi theta delta alpha kappa rho eta upsilon psi
##  [1,] 1  7   310     1    33     1   1 324       2  54
##  [2,] 1  7   320     1    32     1   1 331       2  54
##  [3,] 1  7   314     1    30     1   1 326       2  54
##  [4,] 1  7   317     1    32     1   1 327       2  53
##  [5,] 1  8   319     1    33     1   1 324       2  54
##  [6,] 1  7   317     1    40     1   1 322       2  52
##  [7,] 1  7   313     1    30     1   1 328       2  55
##  [8,] 1  7   319     1    32     1   1 327       2  54
##  [9,] 1  7   312     1    31     1   1 323       2  55
## [10,] 1  7   317     1    31     1   1 329       2  56
## [11,] 1  7   309     1    32     1   1 325       2  55
## [12,] 1  7   313     1    33     1   1 321       2  54
## [13,] 1  7   312     1    35     1   1 318       2  53
## [14,] 1  7   303     1    32     1   1 327       2  53
## [15,] 1  7   304     1    33     1   1 328       2  53
## [16,] 1  7   310     1    31     1   1 321       2  56
## [17,] 1  7   308     1    31     1   1 325       2  56
## [18,] 0  6   312     1    32     1   1 321       2  55
## [19,] 1  7   313     1    33     1   1 321       2  54
## [20,] 1  7   314     1    34     1   1 318       2  53
## [21,] 1  7   317     1    33     1   1 319       2  52
## [22,] 1  7   311     1    32     1   1 312       2  52
## [23,] 1  7   313     1    32     1   1 313       2  52
```

```
## [24,] 1  7   311     1    32     1   1 312       2  53
## [25,] 1  7   316     1    30     1   1 321       2  54
## [26,] 1  7   320     1    33     1   1 316       2  54
## [27,] 1  7   312     1    34     1   1 315       2  52
## [28,] 1  7   321     1    34     1   1 324       2  53
## [29,] 1  7   324     1    33     1   1 321       2  54
## [30,] 1  7   322     1    33     1   1 323       2  53
## [31,] 1  7   319     1    31     1   1 319       2  57
## [32,] 1  7   321     1    31     1   1 324       2  54
## [33,] 1  7   318     1    32     1   1 327       2  55
## [34,] 1  7   320     1    33     1   1 335       2  55
## [35,] 1  7   317     1    35     1   1 323       2  53
## [36,] 1  7   321     1    35     1   1 321       2  53
## [37,] 1  7   328     1    33     1   1 328       2  53
## [38,] 1  7   327     1    33     1   1 325       2  55
## [39,] 1  6   325     1    32     1   1 324       2  55
## [40,] 1  6   321     1    31     1   1 325       2  54
## [41,] 1  7   315     1    32     1   1 325       2  53
## [42,] 1  6   309     1    33     0   1 324       2  52
## [43,] 1  7   314     1    35     1   1 327       2  54
## [44,] 1  7   315     1    32     1   1 326       2  54
## [45,] 1  7   322     1    33     1   1 321       2  54
## [46,] 1  7   314     1    32     1   1 325       2  55
## [47,] 1  7   319     1    31     1   1 324       2  55
## [48,] 1  7   318     1    30     1   1 321       2  53
## [49,] 1  6   315     1    33     1   1 324       2  52
## [50,] 1  7   314     1    34     1   1 328       2  53
```

We then present vital posterior sampling time summary statistics for the 10 key parameters.

```r
round(apply(GibbsStepTimeFixedLfullGP.fast, 2, summary), 2)
```

```
##              z    xi  theta delta alpha kappa rho    eta upsilon   psi
## Min.     73.00 31.00 300.00     1     2  1.00 1.0 307.00    2.00 51.00
## 1st Qu.  82.00 33.00 317.00     1     3  1.00 1.0 326.00    2.00 54.00
## Median   85.00 33.00 325.00     1     3  1.00 1.0 333.00    2.00 55.00
## Mean     87.35 33.32 326.02     1     3  1.03 1.4 334.27    2.02 54.96
## 3rd Qu.  90.00 34.00 334.00     1     3  1.00 2.0 342.00    2.00 56.00
## Max.    260.00 43.00 401.00     2     6  4.00 4.0 403.00    6.00 73.00
```

```r
round(apply(GibbsStepTimeFixedLblock.fast, 2, summary), 2)
```

```
##              z    xi  theta delta alpha kappa  rho    eta upsilon  psi
## Min.     70.00 31.00 300.00     1  2.00  1.00 1.00 309.00    2.00 51.0
## 1st Qu.  79.00 32.00 318.00     1  3.00  1.00 1.00 326.00    2.00 54.0
## Median   81.00 33.00 325.00     1  3.00  1.00 1.00 333.00    2.00 55.0
## Mean     82.53 33.19 326.69     1  2.98  1.01 1.21 334.82    2.02 55.1
## 3rd Qu.  84.00 34.00 334.00     1  3.00  1.00 1.00 342.00    2.00 56.0
## Max.    253.00 42.00 402.00     3  6.00  3.00 3.00 417.00    5.00 72.0
```

```r
round(apply(GibbsStepTimeFixedLsequen.fast, 2, summary), 2)
```

```
##              z    xi  theta delta alpha kappa  rho    eta upsilon   psi
## Min.     68.00 31.00 299.00     1  8.00  1.00 1.00 307.00    2.00 50.00
## 1st Qu.  74.00 32.00 318.00     1  9.00  1.00 1.00 326.00    2.00 54.00
## Median   75.00 33.00 326.00     1  9.00  1.00 2.00 333.00    2.00 55.00
## Mean     76.74 33.17 327.09     1  9.02  1.04 1.62 335.07    2.03 55.13
```

```
## 3rd Qu.  77.00 34.00 335.00     1  9.00  1.00 2.00 342.00    2.00 56.00
## Max.    245.00 44.00 393.00     2 13.00  5.00 5.00 415.00    6.00 72.00
```

```r
round(apply(GibbsStepTimeVaryLjSequen.fast, 2, summary), 2)
```

```
##            u     xi  theta delta  alpha kappa  rho    eta upsilon   psi
## Min.    0.00   6.00 297.00     1  29.00  0.00 1.00 306.00       2 51.00
## 1st Qu. 1.00   7.00 315.00     1  32.00  1.00 1.00 323.00       2 54.00
## Median  1.00   7.00 319.00     1  33.00  1.00 1.00 328.00       2 54.00
## Mean    0.96   7.02 321.41     1  33.26  0.99 1.13 330.12       2 54.73
## 3rd Qu. 1.00   7.00 327.00     1  34.00  1.00 1.00 335.00       2 56.00
## Max.    1.00 148.00 401.00     1 186.00  2.00 3.00 390.00       4 69.00
```

```r
round(apply(GibbsStepTimeFixedLfullGP.fast, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  14.810   1.148  11.308   0.028   0.220   0.231   0.513  11.538   0.224   1.984
```

```r
round(apply(GibbsStepTimeFixedLblock.fast, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  14.404   1.112  11.799   0.028   0.173   0.124   0.413  11.688   0.155   1.966
```

```r
round(apply(GibbsStepTimeFixedLsequen.fast, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  14.191   1.067  11.782   0.014   0.355   0.288   0.527  12.176   0.260   2.004
```

```r
round(apply(GibbsStepTimeVaryLjSequen.fast, 2, sd), 3)
```

```
##       u      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##   0.203   2.040  10.328   0.000   8.447   0.085   0.337  10.418   0.035   1.774
```